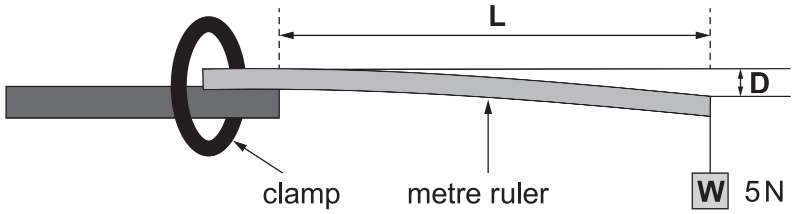
7. Students are investigating how the stiffness of a wooden metre ruler depends on its length. They use the apparatus shown below. The weight, **W**, on the end of the ruler remains constant at 5 N.



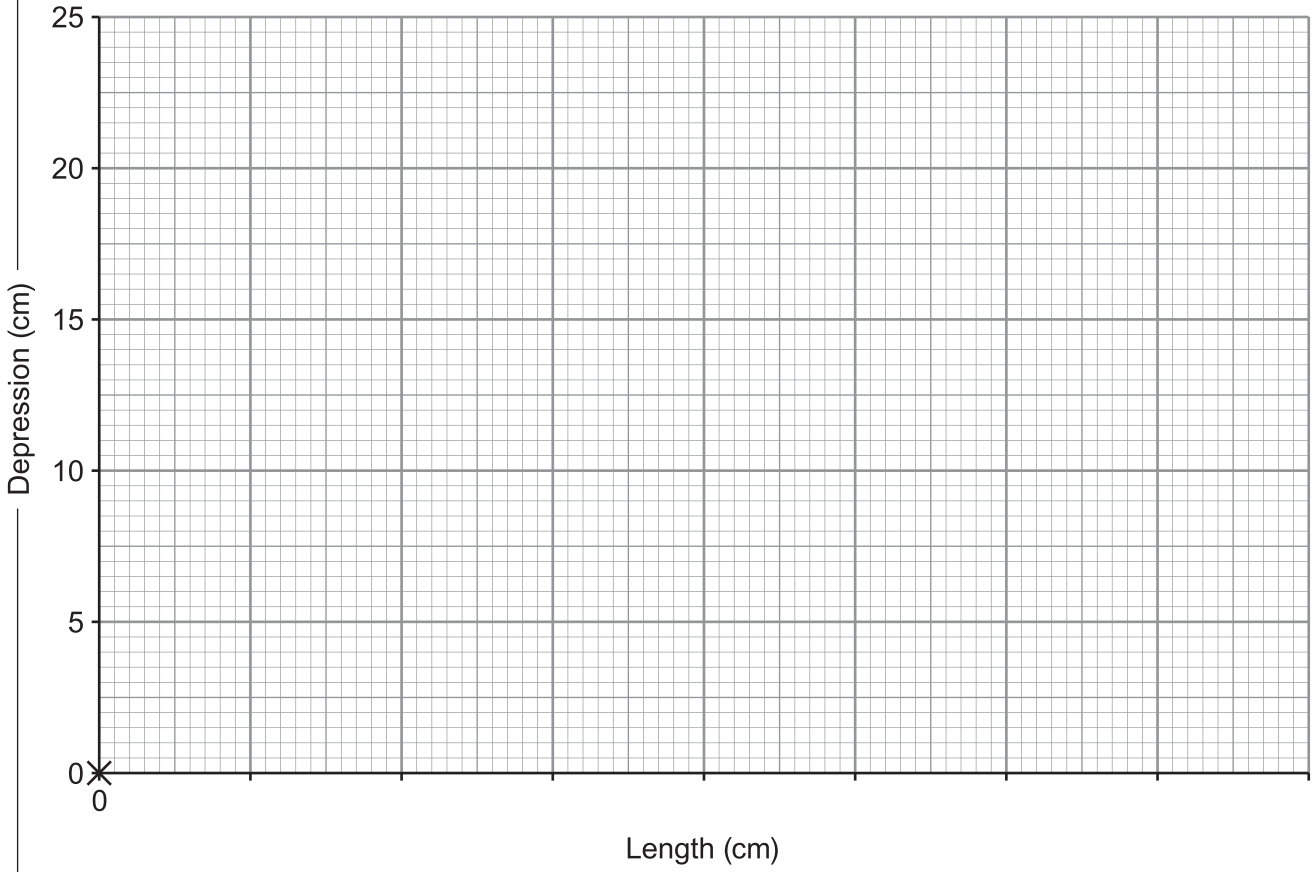
The length, **L**, is varied. For each length, **L**, the depression, **D**, of the end of the ruler is measured.

Their results are shown in the table below.

Length **L** (cm)	Depression **D** (cm)
0	0.0
30	1.0
40	3.0
50	6.0
60	10.0
70	15.0
80	22.0

Examiner only

(a) Plot the data on the grid below and draw a smooth curve. The point (0,0) has already been plotted for you. [4]



(b) It was expected that depression, **D**, would be directly proportional to length, **L**. Explain whether the graph shows a proportional relationship. [2]

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....

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(c) Use information in the table and the equation:

$$\text{stiffness} = \frac{\text{weight}}{\text{depression}}$$

to calculate the stiffness of the ruler at a length of 50 cm. [3]

stiffness = ... N/cm

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